

Technology Stack

Date	15 March 2026
Team ID	xxxxxx
Project Name	Smart Lender
Maximum Marks	2 Marks

Define Your Technology Stack:

- ☐ **Frontend:** HTML5, CSS3
- ☐ **Backend:** Python, Flask
- ☐ **Machine Learning:** Scikit-learn, XGBoost
- ☐ **Data Processing:** Pandas, NumPy
- ☐ **Visualization:** Matplotlib, Seaborn
- ☐ **Data Balancing:** SMOTE
- ☐ **Model Storage:** Pickle (rdf.pkl)
- ☐ **Development Tools:** Anaconda Navigator, PyCharm
- ☐ **Version Control:** Git, GitHub
- ☐ **Deployment Platform:** IBM Cloud

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3	HTML5 and CSS3 are used to develop a simple, responsive, and user-friendly interface. The frontend consists of home.html , predict.html , and submit.html , enabling users to enter applicant details and view loan prediction results.
2	Backend Server-Side	Python 3.x, Flask	Python is used for backend development and machine learning integration. Flask acts as a lightweight web framework that manages routing, processes HTTP POST requests, validates user input, loads the trained model, performs predictions, and returns the result to the user interface.
3	Database/ Data Storage	CSV Dataset, Pickle (.pkl) Model Storage	The loan applicant dataset is stored as a CSV file for model training. The trained machine learning model is serialized and stored as an rdf.pkl file using the Pickle library, allowing fast loading during prediction without retraining.
4	Cloud/ Hosting/ Deployment	IBM Cloud	The Smart Lender application is deployed on IBM Cloud, providing reliable cloud hosting, browser-based accessibility, scalability, and simplified deployment for users without requiring local installation.
5	Version Control& CI/CD	Git, GitHub	Git is used for source code version control, while GitHub provides repository management, collaboration among team members, code backup, and project version tracking throughout development.
6	Third-Party APIs/Other Tools	Anaconda Navigator, PyCharm, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn,	Anaconda manages Python environments and packages. PyCharm is used for development and debugging. Pandas and NumPy perform data manipulation and numerical computations.

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		imbalanced-learn (SMOTE), Pickle	Matplotlib and Seaborn generate visualizations for exploratory data analysis. Scikit-learn provides preprocessing utilities and machine learning algorithms such as Decision Tree, Random Forest, and KNN. The imbalanced-learn library implements SMOTE to balance the dataset. Pickle serializes the trained model for deployment within the Flask application.